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Feasibility of asymmetrical flow field-flow fractionation as a method for detecting protective antigen by direct recognition of size-increased target-captured nanoprobes

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ABSTRACT

Asymmetrical flow field-flow fractionation (AF4) was evaluated as a potential analytical method for detection of a protective antigen (PA), an *Anthrax* biomarker. The scheme was based on the recognition of altered AF4 retention through the generation of the size-increased Au nanoparticle probes as a result of PA binding, in which a PA-selective peptide was conjugated on the probe surface. In the visible absorption-based AF4 fractograms, the band position shifted to a longer retention time as the PA concentration increased due to the presence of probe bound with PAs. The shift was insignificant when the concentration was relatively low at 84.3 pM. To improve sensitivity, two separate probes conjugated with two different peptides able to bind on different PA epitopes were used together. The band shift then became distinguishable even at 84.3 pM of PA sample. The formation of larger PA-probe inter-connected species using the dual-probe system was responsible for the enhanced band shift. In parallel, the feasibility of surface-enhanced Raman scattering (SERS) as a potential AF4 detection method was also evaluated. In the off-line SERS fractogram constructed using fractions collected during AF4 separation, a band shift was also observed for the 84.3 pM PA sample, and the band intensity was higher when using the dual-probe system. The combination of AF4 and SERS is promising for the detection of PA and will become a potential tool if the reproducibility of SERS measurement is improved.

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1. Introduction

Studies on disease diagnosis through the detection of a relevant protein biomarker, especially using diverse nanostructures based on noble metals and quantum dots made of semiconductor materials, have been extensively reported [1–9]. There are several advantages of employing nanostructures in the analyses, such as sensitivity enhancement and multiplexed detection [3–9]. Probes prepared by conjugating target-specific receptors such as antibodies or peptides on the surface of nanoparticles or quantum dots are then used for selective detection of relevant biomarkers. When a nanostructure-based probe captures a target protein biomarker, its overall size is expected to increase. Therefore, a method capable of recognizing the increased size of the target-captured probe, such as field-flow fractionation (FFF) [10–14], is potentially useful since it could make the analysis reliable by directly observing the analyte-induced physical change in the probe and by eliminating sample-handling steps, such as washing of target-uncaptured probes, which is commonly needed in a normal plate-based immunoassay [15]. In addition, the immune-reaction is kinetically favored when nano- or micrometer-sized beads are used as an assay substrate compared to the conventional microtiter plate assays [15–19].

FFF is a family of separation techniques useful for size characterization of particles in the range from a few nm to approximately 100 μm [20–22], and its separation is performed by applying an external field to the sample flowing in a hollow channel without a solid stationary phase, relieving problems of non-specific protein adsorption or structural deformation on surface. In the case of FFF-based immunoassay for protein detection, a few studies utilizing micrometer-sized beads coated with capture antibody selective to target analytes have been reported [15–19], such as the detection of horseradish peroxidase (HRP) using chemiluminescence [15–18] and the measurement of protein A by combining flow cytometry

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[19]. In these studies, micron-sized polystyrene (PS) beads were used since they are readily separable in a FFF channel, chemically stable, and facile for use in surface functionalization.

Another way of utilizing FFF for protein detection is the direct recognition of the size-increased target-bound particle and use of an altered fractogram to measure target concentration. This method is simple since a mixture of target and probe particle is injected into an FFF channel for measurement without further sample treatments. Moreover, size increase of the particles can be easily recognized via migration time shift if the particle is directly detectable. To realize this strategy, several important issues should be considered. First, the size of the probe capturing a target protein needs to be properly selected in consideration of the relative size difference between bare probe and target-captured probe. As known size of a protein is usually a few nanometers [23], the use of a nano-sized probe would be advantageous because the relative size change is larger than conventionally adopted micron-size particles. Also, the probe needs to be quickly detected upon FFF elution. Considering these criteria, the above-mentioned noble metal nanoparticles and quantum dots seem to be good FFF probe candidates since their sizes are usually in the range of nanometers, and they have spectroscopic properties, such as visible absorption (color) and fluorescence [5–9], allowing fast optical detection.

In this study, we explored the feasibility of asymmetrical flow FFF (AsFlFFF, AF4) as an analytical method for the detection of a protective antigen (PA), an *Anthrax* biomarker [24–26], by recognizing the altered feature of the AF4 fractogram induced by the size increase of PA-captured Au nanoparticles (AuNPs). In AF4, the hydrodynamic diameter, d_H, of a particle can be determined directly from its measured retention time, t_r, according to

$$d_H = \frac{2kTV^o}{\pi\eta w^2 F_c t^o} t_r \quad (1)$$

where k is the Boltzmann constant, T is the absolute temperature, V^o is the channel void volume, η is the viscosity of the carrier liquid, w is the channel thickness, F_c is the volumetric crossflow rate, and t^o is the channel void time [27–30]. Eq. (1) shows that d_H is proportional to t_r, so that an AF4 fractogram is a direct representation of the size distribution of the sample that can be readily transformed to a size distribution. Therefore, increased retention time is indicative of the presence of a species with larger hydrodynamic diameter, such as PA-captured AuNPs.

Initially, ~60 nm AuNPs were synthesized, and a peptide that can selectively bind to PA was conjugated onto the AuNP surface using a linker molecule (5,5′-dithiobis(succinimidyl-2-nitrobenzoate), DSNB), as shown in Fig. 1. The peptide-conjugated AuNP probes were added to PA samples at various concentrations, and the mixture solutions were injected into an AF4 system equipped with a UV/Visible detector. Then the changes in AF4 fractograms with PA concentration were examined. When two probes conjugated with different peptides capable of binding on different PA epitopes were used, larger species such as PA bound with both probes and inter-connected multiple probes were formed. These complexes were more perceptible in an FFF system, thus further improving the sensitivity of PA detection. Dual probes separately conjugated with two different PA-selective peptides (Fig. 1) were used to measure the same PA samples, and the potential sensitivity improvement was examined.

AuNP is a widely adopted SERS material and thus SERS-based detection of FFF effluent is feasible since the linker molecule (DSNB) on the Au surface simultaneously operates as an effective Raman reporter [31]. SERS has been widely employed for analysis of diverse samples due to its ultra-sensitivity [32,33]. Therefore, synergetic combination of both methods, the capability of FFF for recognizing probe size increase and the high sensitivity of SERS detection, is worthwhile to explore. For the feasibility study, AF4 fractions were collected during an FFF elution, and Raman spectra of each fraction were acquired to construct off-line SERS-based fractograms. The positive potential and challenges of SERS-based detection are subsequently discussed.

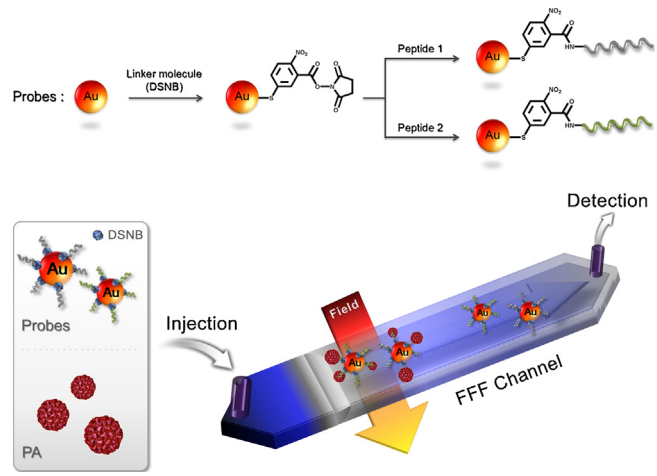

Fig. 1. Schematic descriptions of the preparation of peptide-conjugated AuNP probes (top) and the overall PA detection scheme using FFF (bottom).

2. Experimental

2.1. Preparation of AuNP probes

All reagents used in the study were of the highest available grade, purchased from Sigma–Aldrich, except FL-70, and used as received. FL-70, a dispersing agent, was obtained from Fisher Scientific (Fair Lawn, NJ, USA). First, 15 nm AuNPs were synthesized through a well-known citrate reduction method [34]. The average size of the synthesized AuNPs was 14.6 ± 1.4 nm, with a maximum absorption at 520 nm (concentration: 9.27×10^{12} particles/mL). Then, the 15 nm AuNPs were grown to a diameter of 30 or 60 nm through the surface-catalyzed reduction of Au^{3+} by NH_2OH, the so-called seed mediated growth method [35,36]. The average sizes of two synthesized AuNPs were 32.0 ± 3.3 (1.51×10^{11} particles/mL) and 59.6 ± 6.3 nm (2.29×10^{10} particles/mL), with maximum absorption observed at 525 and 530 nm, respectively. The size distribution of AuNPs was determined by the measured sizes of dispersed particles in SEM images (obtained by Hitachi S-4800) using an image analysis program (Image J, National Institutes of Health, USA).

To prepare the AuNP probes, 25 mL of the 60 nm AuNP solution was mixed with 25 mL of water, and then the pH of the solution was adjusted to 8.5 using 1 M NaOH or HCl solution. Then, 2 mL of 50 mM borate buffer was added to the pH-adjusted solution. To conjugate a linker (5,5′-dithiobis(succinimidyl-2-nitrobenzoate), DSNB) to the AuNP surface, 125 μL of 1 mM DSNB solution was added to the AuNP solution. This procedure generated a thiolate monolayer of DSNB on the AuNP surface, and the succinimidyl terminus in the DSNB-derived coating formed an amide linkage with a primary amine of the peptide (Fig. 1), as previously reported [31]. Two different AuNP probes were prepared by coupling two peptides with different amino acid sequences (peptide 1 sequence: HKHAHNYRLPASGGKK (K_d = 28.0 ± 3.6 nM), peptide 2 sequence: NAYKHHHPPVFYGKGK (K_d = 4050.8 ± 18.6 nM)). The peptides (synthesized by Peptron, Daejeon, Korea) bind to different epitopes on PA, as previously reported [37–39]. Either single or dual probes were mixed with a PA sample solution over three hours at room temperature before injection into the AF4 channel.

2.2. Acquisition of AF4 fractograms

The AF4 system consisted of an Eclipse AF4 short channel (Wyatt Tech., Europe GmbH, Dernbach, Germany) assembled with a regenerated cellulose membrane with the cutoff-molecular weight of 10 kDa (Millipore, Bedford, USA) and a 350-μm-thick Mylar spacer. The channel geometry was trapezoidal with a length of 17.3 cm. The channel widths at an inlet and outlet were 1.6 and 0.3 cm, respectively. The channel thickness determined based on the retention time of ferritin was 250 μm, and the focusing time was 50 s at a flowrate of 1.5 mL/min in all experiments.

The carrier liquid was composed of 0.1% FL-70 and 0.02% NaN_3 (a bactericide) in purified water prepared using a Milli-Q plus purification system (Millipore Co. Ltd., Billerica, USA, resistance = 18.2 MΩ/cm). The solution was vacuum-filtered through a 0.45 μm membrane filter (Membrane Solutions, Plano, USA) before use and introduced through the AF4 channel using a HPLC pump (LC-20AD, Shimadzu Corp., Kyoto, Japan) with an OptiFlow 1000 liquid flowmeter (Agilent Technologies, Palo Alto, CA, USA). Eluted samples were monitored by a UV/Vis detector (Model 500, Chrom Tech, MN, USA). For detection of 15, 30, and 60 nm AuNPs, the detector wavelength was set to 520, 525, and 530 nm, respectively. The injected sample volume was 50 μL, and all AF4 experiments were performed at room temperature. Each sample was vortexed for 10 s using a Genie 2 (A. Daigger & Co., Inc., Chicago, IL, USA) before injection into the AF4 channel.

2.3. Acquisition of Raman spectra of AF4 fractions

AF4 fractions were continuously collected for 30 s, and Raman spectra of each AF4 fraction were obtained using a Raman microscope (Kaiser Optical System, Inc., Ann Arbor, MI, USA) equipped with a diode laser (λ_{ex} = 785 nm, power = 44 mW), a thermoelectrically cooled (−40 °C) charge-coupled detection (CCD) detector, and holographic grating (Kaiser Optical Inc., Ann Arbor, MI, USA). A sample was positioned on a microscope stage, and the laser beam was focused using an objective lens (10×/0.25NA) in order to collect spectra at room temperature with a resolution of 4 cm^{-1}. For each AF4 fraction, 10 SERS spectra (laser exposure time: 1 s) were collected by randomly illuminating the sample. The intensities of the DSNB peaks at 1340 cm^{-1} were used to construct off-line SERS-based fractograms.

3. Results and discussion

3.1. Separation of AuNPs with different sizes

The capability of AF4 for separation of 15, 30, and 60 nm AuNPs was evaluated. For this, three binary mixtures of 15/30 nm, 15/60 nm, and 30/60 nm AuNPs were prepared and subjected to AF4 separation. AF4 fractograms of these mixtures are shown in Fig. 2(a), where the *x*-axis represents both retention time (bottom) and measured hydrodynamic diameter (top). The UV/Vis detector wavelength was set at 530 nm. The outflow rate (F_{out}) and cross-flow rate (F_c) were 0.78 and 0.77 mL/min, respectively. As shown, the mixtures were well separated by AF4, and the measured hydrodynamic diameters are in good agreement with the average sizes of AuNPs determined by image analysis, as in Section 2. The elution bands of 60 nm AuNPs were broader than those of 15 and 30 nm AuNPs due to a combination of increased band broadening (caused by higher retention) and larger particle size distribution.

Next, the variation in UV/Vis detector response with AuNP concentration was examined. For this, 60 nm AuNP samples with five different concentrations (2.29×10^{10}, 1.15×10^{10}, 0.58×10^{10}, 0.29×10^{10}, and 0.15×10^{10} particles/mL) were prepared, and their

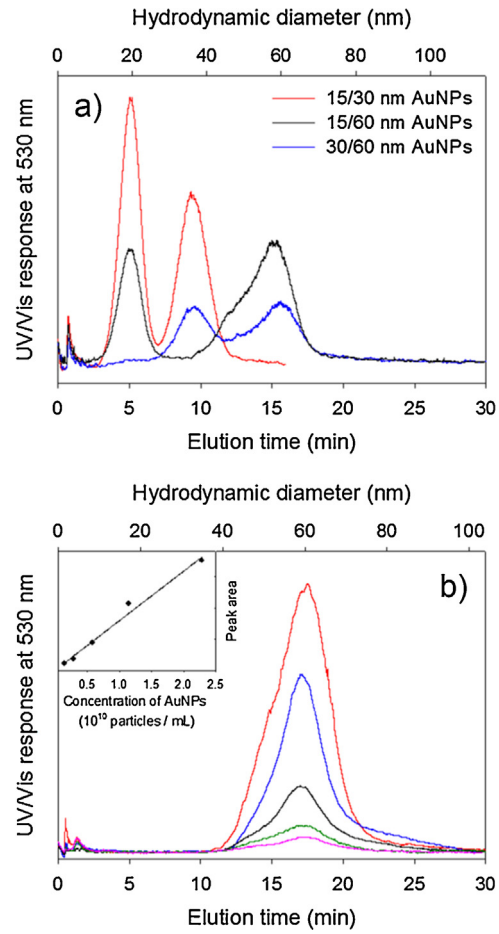

Fig. 2. (a) AF4 fractograms of three binary mixtures of 15/30 nm (red), 15/60 nm (black), and 30/60 nm AuNPs (blue) (F_{out}: 0.78 mL/min, F_c: 0.77 mL/min). (b) AF4 fractograms of 60 nm AuNPs at five different concentrations (2.29×10^{10} (red), 1.15×10^{10} (blue), 0.58×10^{10} (black), 0.29×10^{10} (green), and 0.15×10^{10} particles/mL (pink)) (F_{out}: 0.84 mL/min, F_c: 0.67 mL/min). UV/Vis detector wavelength was set at 530 nm. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

AF4 fractograms are shown in Fig. 2(b). The UV/Vis detector wavelength was 530 nm, and F_{out} and F_c were 0.84 and 0.67 mL/min, respectively. The inset shows a plot of band areas vs. particle concentration with the regression line. In the tested range, the detector response gradually increases as the AuNP concentration increases. Fig. 2(b) shows that AF4 is sufficient to detect AuNP concentration of 0.15×10^{10} particles/mL.

3.2. Variation in AF4 elution upon capture of PA by AuNP probe

When the peptide-conjugated AuNP probe binds with PA, the overall size of the probe becomes larger due to PA binding, and its AF4 retention is expected to increase. A solution of 60 nm AuNP probes (concentration: 1.15×10^{10} particles/mL) conjugated with peptide 1 was mixed with a 100 nM PA sample. The AF4 fractograms of the AuNP probes before (blue) and after (red) mixing are shown in Fig. 3(a). The UV/Vis detector wavelength was set at 530 nm, and the F_{out} and F_c were 0.22 and 1.31 mL/min, respectively. The size of PA is approximately 100 × 50 × 30 Å [24], and its binding with the probe will increase the overall size of the probe complex, resulting in an increase in AF4 retention, as shown in Fig. 3(a). Also, the AF4 elution band increased in width probably due to co-existence of the bare probes and PA-captured probes. Probes bound to more than one PA (designated as Probe-PAs) could also exist. The formation of Probe-PAs is probably advantageous to enhance the detection sensitivity

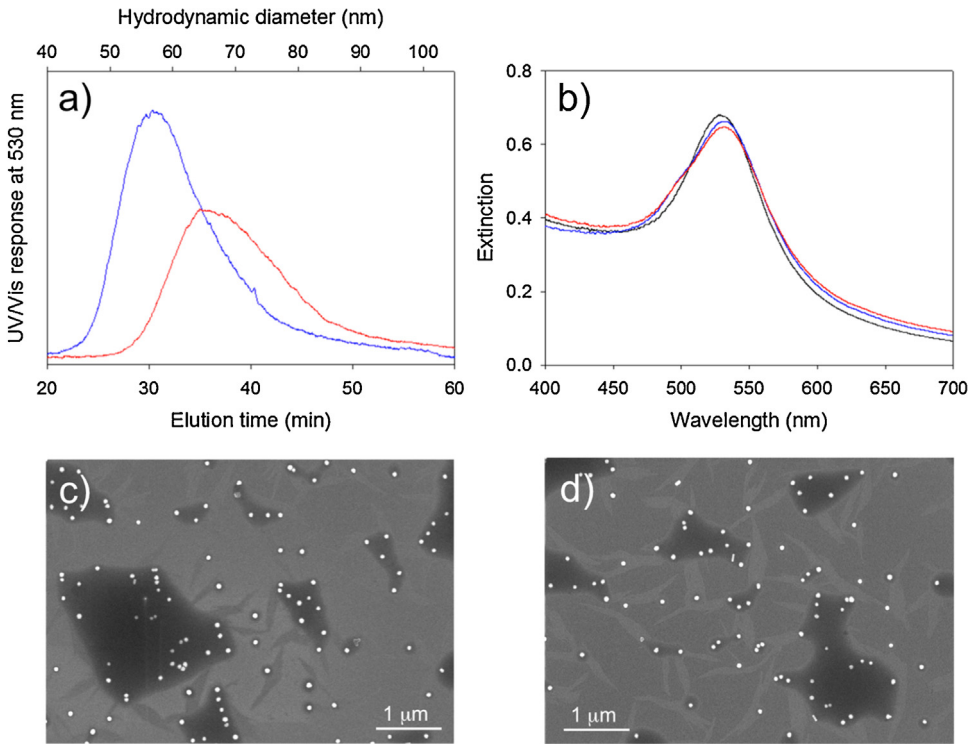

Fig. 3. (a) AF4 fractograms of AuNP probe before (blue) and after (red) mixing with 100 nM PA (F_{out}: 0.22 mL/min, F_c: 1.31 mL/min). (b) Extinction spectra of bare AuNPs (black), bare AuNP probes (blue), and PA-captured AuNP probes (red). SEM images of bare probes (c) and PA-captured probes (d) eluted from the AF4 channel are also shown. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

by the increased size difference; while, it would also result in band broadening if the sizes of formed Probe-PAs are inconsistent.

Other than PA binding, unwanted events such as probe aggregation can also change the size of probes. Extinction spectra of the bare AuNPs, bare probes, and PA-captured probes are presented in Fig. 3(b). The characteristic absorption bands of 60 nm AuNPs centered at 530 nm are clearly observed for all three samples, and their band positions are nearly unchanged. This suggests there was no noticeable probe aggregation. Next, SEM images of the bare probes and PA-captured probes eluted from the AF4 channel were examined, as shown in Fig. 3(c) and (d), respectively. In both SEM images, AuNP probes are well dispersed without noticeable aggregation, and there is no significant difference in the spread of AuNPs between these two images. The observation of extinction spectra and SEM images confirm that the shift in AF4 retention shown in Fig. 3(a) is predominantly attributed to the increased size of the probes due to binding with one (Probe-PA) or more PAs (Probe-PAs).

It is expected that AF4 retention will continue to increase with increasing PA concentration since higher PA concentration results in an increase in the number of PAs bound to a probe. PA samples of four different concentrations (84.3 pM, 0.843 nM, 8.43 nM, and 84.3 nM) were mixed with the probe solution, and their AF4 fractograms are shown in Fig. 4(a) along with that of a control sample (probe only). The points denote absorptions at 530 nm, and the lines are the corresponding Weibull-fitted fractograms. A Weibull distribution was empirically the most descriptive to explain the variation in data points and to determine exact band positions.

The AF4 elution time was plotted vs. PA concentration, as shown in Fig. 4(b). The error bars were calculated using the results from three independent measurements. No significant difference in AF4 retention from that of the control was observed at the PA concentrations of 84.3 pM and 0.843 nM. It seems the size differences between bare probes and PA-captured probes are too small to be differentiated by AF4. At 8.43 nM, however, AF4 retention was increased significantly, and even more at 84.3 nM. The increase in AF4 retention with increasing PA concentration indicates that relatively larger species such as more PA-captured Probe-PAs are formed at higher PA concentrations, as expected.

It is known that peptides with different sequences can bind to various epitopes on PA [37–39]. PA (PA83) is composed of two blocks, PA63 and PA20 [24]. Peptide 1 and peptide 2 are selective to PA63 and PA20 with K_d values of 28.0 ± 3.6 nM and 4050.8 ± 18.6 nM, respectively [38]. When the second probe conjugated with peptide 2 (designated Probe (2)) was mixed with the 100 nM PA sample, the shift in AF4 retention to a longer retention time from that of a control sample, as observed with Probe (1), was apparent (relevant AF4 fractograms are not presented here). This result indicates that the species of Probe (2)-PA and Probe (2)-PAs are also formed when Probe (2) is employed.

Next, both Probe (1) and Probe (2) were used together, and the results are shown in Fig. 4(c) and (d). The simultaneous use of two probes resulted in formation of new species in which both probes are inter-connected via PA, such as PAs-Probe (1)-PA-Probe (2)-PAs and Probe (1)-PA-Probe (2). The formation of these larger inter-connected complexes is helpful to improve sensitivity. Fig. 4(d) shows the variation in AF4 elution time with PA concentration. Under these conditions, the shift (toward longer retention) in AF4 retention was distinguishable even at low PA concentrations (84.3 pM and 0.843 nM). The results show that the increase in AF4 retention is greater in a dual-probe system (Fig. 4(c) and (d)) than in a single-probe system (Fig. 4(a) and (b)) for the measurement of equivalent PA concentrations, due to the formation of larger inter-connected PA-probes.

3.3. Feasibility of SERS measurement as a detection method for AF4

Since AuNP can generate a SERS signal from adsorbed molecules, SERS measurement is a feasible alternative detection method for

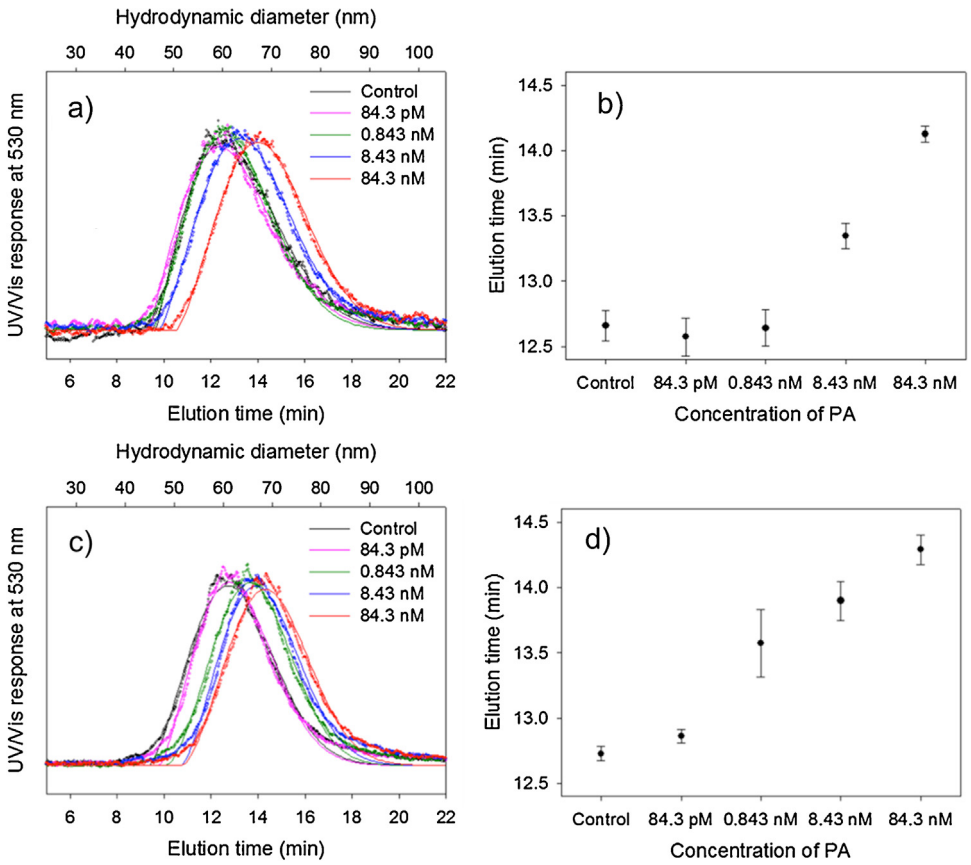

Fig. 4. AF4 fractograms acquired from PA samples of various concentrations (84.3 pM, 0.843 nM, 8.43 nM, and 84.3 nM) (a), and a plot of elution time vs. PA concentration (b) when a single probe is used for the measurements. AF4 fractograms (c) and variation in elution time according to PA concentration change (d) when using dual probes are also shown. F_{out} and F_c were 1.0 and 1.5 mL/min, respectively, in both cases.

AF4. Therefore, the adoption of DSNB as a linker was to consider SERS measurement in parallel since it can also function as a Raman reporter due to the high Raman cross-section provided by the NO_2 group in its structure [31]. For the preliminary evaluation, both DSNB-conjugated 15 nm and 60 nm AuNPs (DSNB-15NPs and DSNB-60NPs, respectively) were prepared. The 50:50 mixture of both samples was then injected into the AF4 system, and Raman spectra of AF4 fractions periodically collected over a period of 30 s were measured to construct an off-line SERS fractogram. The spectra were collected by random laser illumination of 10 different spots on each sample, as described earlier.

Fig. 5 shows the fractogram of the 50:50 mixture of DSNB-15NPs and DSNB-60NPs constructed using Raman intensity at 1340 cm^{-1} (the strongest DSNB peak). The error bars correspond to the standard deviations of 10 separate measurements for a given fraction. The inset shows the SERS spectrum of DSNB in the 1640–600 cm^{-1} range, and the 1340 cm^{-1} band is marked with an asterisk. Two separated bands corresponding to 15 and 60 nm AuNPs are apparent in the SERS-based AF4 fractogram, with the elution times of the bands similar to those of the UV/Vis-based AF4 fractogram (Fig. 2(a)). As shown in Fig. 5, the band intensity of DSNB-60NPs is much higher than that of DSNB-15NPs since the SERS efficiency of 60 nm AuNPs is greater than that of 15 nm AuNPs, as previously reported [40,41].

To examine the intensity variation of the AF4 band with changing AuNP concentration, a DSNB-60NPs sample was prepared by mixing a solution of 60 nm AuNPs (2.29×10^{10} particles/mL) with a DSNB solution (final concentration: 100 μM). The DSNB-60NPs sample was then serially diluted with water by one-to-one volume ratio up to 4 times, and each sample was measured using AF4.

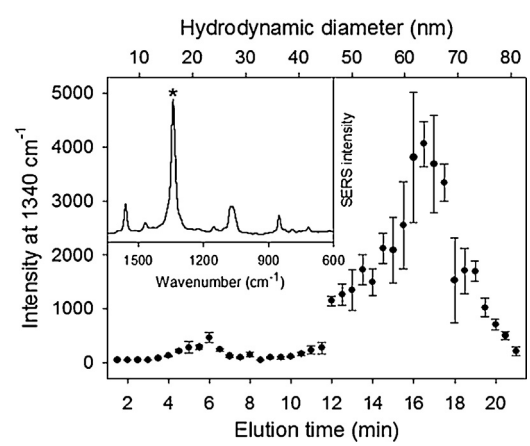

Fig. 5. Off-line SERS-based AF4 fractogram of a 50:50 mixture of DSNB-15NPs and DSNB-60NPs constructed using the DSNB band intensity at 1340 cm^{-1}. The error bars correspond to the standard deviations of 10 separate measurements for a given fraction. The inset shows a SERS spectrum of DSNB in the 1640–600 cm^{-1} range, and the 1340 cm^{-1} band is marked with an asterisk. F_{out} and F_c were 0.78 and 0.77 mL/min, respectively.

Fig. 6 shows SERS-based AF4 fractograms of the DSNB-60NPs and serially diluted DSNB-60NPs samples. The dots are SERS intensities at 1340 cm^{-1}, and the lines are results of Lorentzian-based data fitting. The band intensities decrease as the DSNB-60NPs concentration decreases, as shown in Fig. 6(a)–(e). Fig. 6(f) shows a plot of the maximum SERS intensity vs. concentration of DSNB-60NPs, illustrating that the variation of SERS intensity is linear in the tested range.

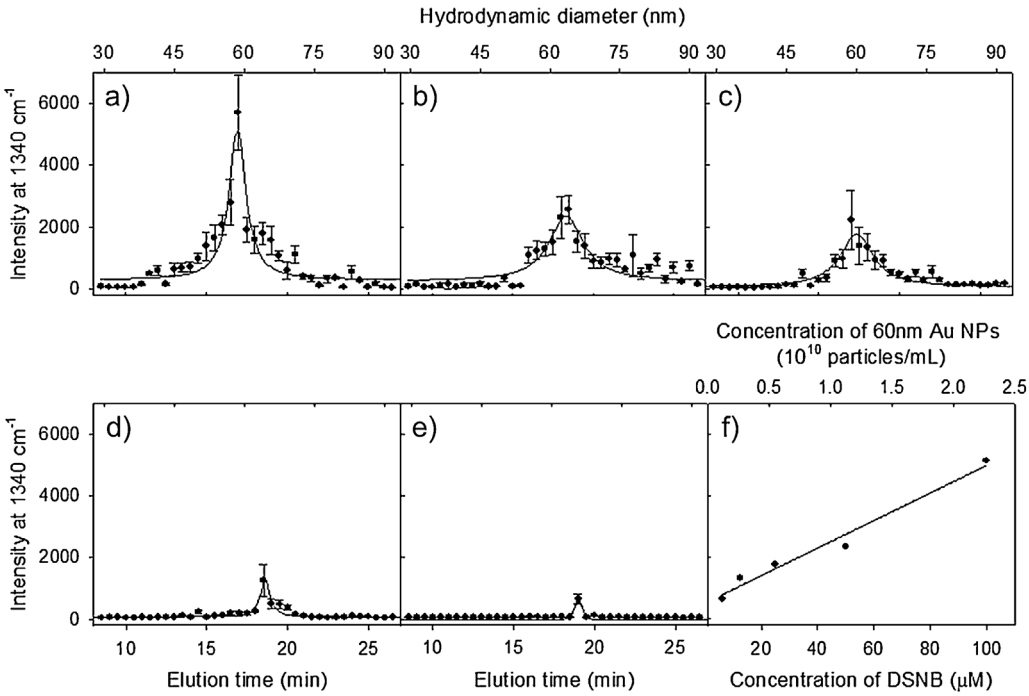

Fig. 6. SERS-based AF4 fractograms of DSNB-60NPs (solution of 60 nm AuNPs (2.29×10^{10} particles/mL) mixed with DSNB (final concentration: 100 μM) (a) and consecutively diluted DSNB-60NPs samples (b)–(e). The dots represent the intensities of the 1340 cm^{-1} peak, and the lines are resultant Lorentzian-based data fit. The variation in peak intensities with DSNB-60NPs concentration is also shown (f). F_{out} and F_c were 0.84 and 0.67 mL/min, respectively.

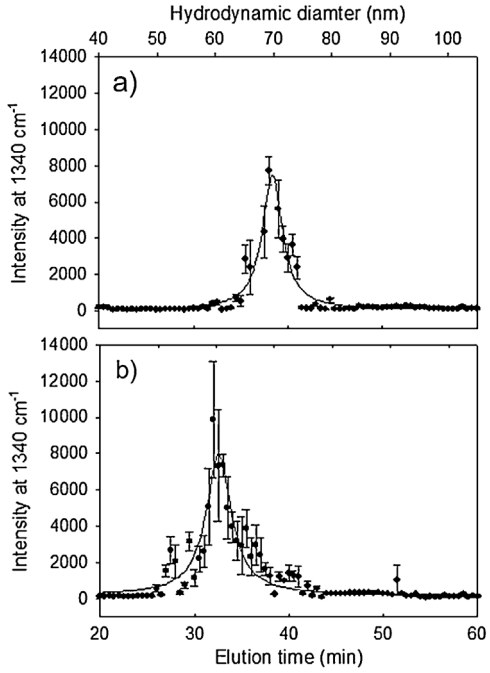

Fig. 7. SERS-based AF4 fractograms of Probe (1) mixed with the 100 nM PA sample (a) and control sample (b). F_{out} and F_c were 0.22 and 1.31 mL/min, respectively.

With the use of Probe (1), a 100 nM PA sample was measured, and the corresponding Lorentzian-fitted SERS-based AF4 fractogram was obtained, as shown in Fig. 7(a). Three outlier fractions that showed no SERS signal were excluded before the fitting. The fractogram of a control sample is shown in Fig. 7(b). In comparison with that of the control sample, the AF4 elution band was shifted toward a longer retention time, as observed earlier via UV/Vis detector. The band shift is due to the generation of Probe (1)-PA and Probe (1)-PAs species, as described previously.

The measurement of the 84.3 pM PA sample (the lowest concentration tested in this study) using single (Probe (1))- and dual (Probe (1) and Probe (2))-probe methods was attempted with SERS detection. Fig. 8(a) and (b) shows SERS-based AF4 fractograms of the single- and dual-probe methods, respectively. The fractogram of a control sample is also shown (c). The dashed lines in each figure indicate the position of the AF4 band of the control sample. When only a single probe was used, no shift in band position compared to the control was also observed in the SERS fractogram as well as UV/Vis-based fractogram. With dual probes, a band shift is distinguishable in the SERS-based fractogram due to the presence of PA-probe inter-connected species, and its intensity is higher than that of the control. The increased intensity would be attributed to the enhancement of SERS intensity resulting from the closer arrangement of probes in the inter-connected species, while it was not observed in the corresponding UV/Vis-based fractogram (Fig. 4).

Although SERS-based AF4 detection for PA is feasible, as shown in Figs. 7 and 8, several issues are necessary to critically be addressed in order to make it a reliable method. First, the reproducibility of SERS measurement needs to be secured for quantitative analysis. For example, the magnitudes of error bars for each measurement are fairly large, which negatively influence the accuracy of SERS-based detection. Second, an on-line SERS detection for AF4 is required for practical applications, which may alleviate most of the above-mentioned problems (i.e., fluctuation in SERS intensity). One of the most feasible approaches is the hyphenation of a microfluidic device at an end of the AF4 channel. This coupling will allow AF4 effluent flow through a narrow channel with minimized flow dispersion, thereby providing a proper platform for consistent on-line SERS detection. Also the design of an illumination chamber of optimal size and shape, which holds the proper number of probes for averaging individual SERS signals in order to minimize intensity variation without losing separation efficiency, needs to follow. A large illumination chamber is advantageous for the acquisition of

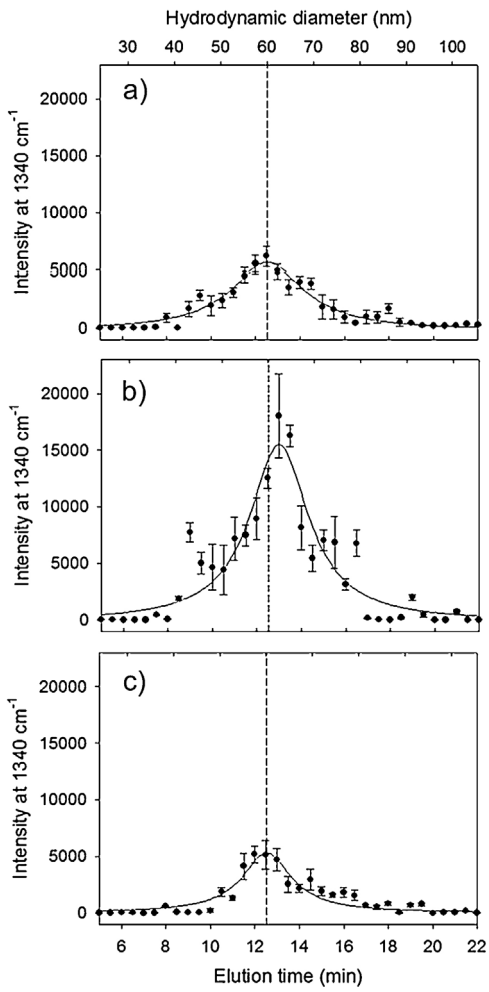

Fig. 8. SERS-based AF4 fractograms obtained with 84.3 pM PA using single (Probe (1)) (a) and dual probes (b). The control fractogram is also shown (c). The dashed lines indicate the AF4 band position of the control sample. F_{out} and F_c were 1.0 and 1.5 mL/min, respectively.

AF4 fractograms with minimal intensity fluctuation, although such a large chamber degrades the resolution of AF4 separation due to band broadening.

4. Conclusions

The feasibility of AF4 for PA detection through the direct recognition of the increase in AuNP probe size upon PA binding has been demonstrated. When two probes that can bind to different PA epitopes were used together, larger and diverse inter-connected PA-probes that are more amenable to AF4 separation were formed, thereby improving sensitivity. As an AF4 detector, SERS-based measurement offers high potential along with conventional detectors such as UV/Vis detector; however, its reproducibility needs to be improved for practical analytical usage. Nonetheless, we think that the combination of AF4 and SERS-based detection is a promising analytical strategy for detection of protein biomarkers for diagnosis of diseases along with a conventional immunoassay-based analysis. Another way to improve detection sensitivity is the generation of larger inter-connected species, such as PA-mediated networked aggregates. To realize this, optimal experimental conditions, such as the amount of peptide on a probe, size of a probe relative to the size of the target protein, and incubation temperature/time, must be determined. Efforts to ameliorate the reproducibility of on-line SERS measurement by employing a microfluidic-based platform and to find a strategy for generating larger networked aggregates are on-going.

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